

PRIMARY SCIENCE

TOPICAL REVISION

ANIMALS



SUMMARIZED SCIENCE REVISION NOTES

ANIMALS

Living things have ways which enable them to continue living called characteristics.

CHARACTERISTICS OF ANIMALS

1. **Feeding** - It is taking food into the body
 - They feed to grow, get energy and be healthy.
 - There are different feeding habits. They include:
 - a) Carnivores, which feed on, flesh
 - b) Herbivores, which feed on plants.
 - c) Omnivores, which feed on both plants and animals.
2. **They grow** - It is changing from young animals to mature ones.
Animals increase in size.
3. **Movement** - It is the action of changing position or location.
They move in order to;
 - Look for shelter.
 - Look for food.
 - Run away from enemies.
 - Escape harsh weather conditions.
 - Look for partners.
 - a. They move by:-
 - Walking
 - Flying
 - Swimming
 - Crawling
4. **Reproducing** - They reproduce by giving birth to young ones or laying eggs
Which hatch to young animals.
5. **Reacting to changes**- All animals react to changes in environment eg cold, rain,
Heat, external threats
6. **Removing waste products** - Animals get rid of waste materials through excretion.
7. **Death**- All animals eventually die.
Their bodies decay and decompose to form humus.

CLASSIFYING FARM ANIMALS

They are classified according to products and services they provide.

(a) POULTRY

- They are domestic birds kept for meat and eggs.
- They include ducks, geese, chickens, turkeys.
- Those chickens kept for meat are called broilers or table birds.
- The other type kept for eggs are called layers.

(b) CATTLE

- They include bulls and cows.
- Those cattle reared for milk are called dairy cattle's and those reared for beef are called cattle.
- Dairy cattle have wedge shaped bodies.
- Beef cattle are block shaped.

(c) SHEEP

- It produces wool and meat called mutton.

(d) CAMELS

- Mainly found in dry areas.
- It is kept for milk, meat and for carrying loads.

(e) DOGS

- Kept for security.

(f) HORSES

- They help pull carts and in sports.

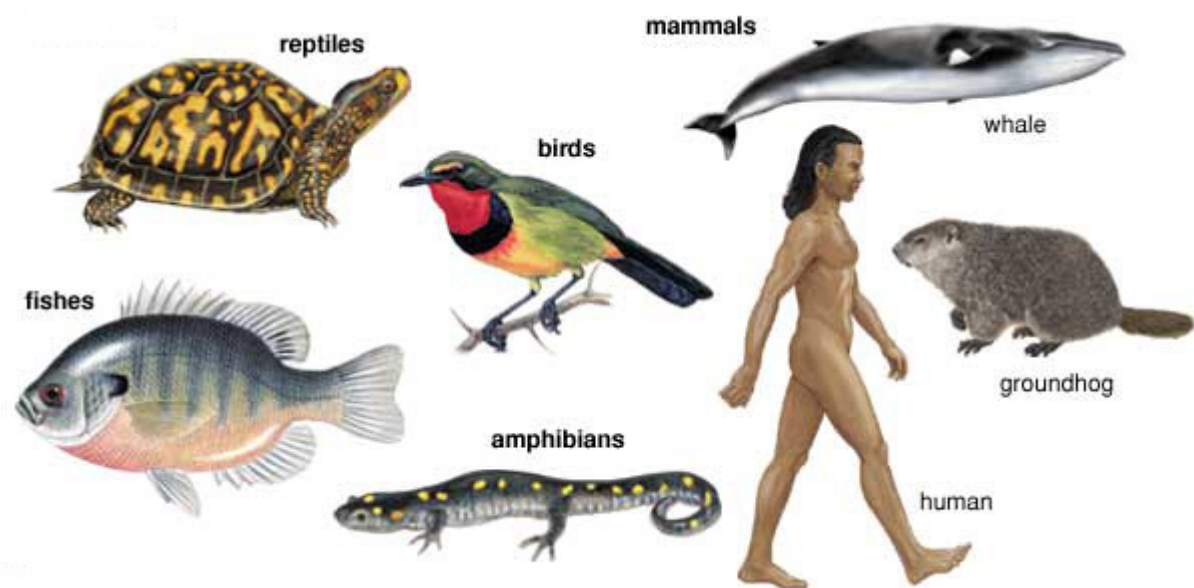
(g) GOATS

- Mainly kept for milk, mohair and mutton,
- Goats kept for milk production are called dairy goats.

CLASSIFICATION OF ANIMALS

- Animals are classified into two main groups vertebrates and invertebrates.
- Vertebrates have backbones and the invertebrates.
- Have no backbones.

VERTEBRATES:



They are divided into cold blooded and warm blooded.

WARM BLOODED (Constant body temperature)

They are divided into two mammals and birds.

MAMMALS:

Characteristics

- They have mammary glands. Female mammals produce milk for their young ones.
- They breath through lungs.
- Most live on dry land and some in water e.g. Whales and Dolphins.
- Most give birth to live young ones except duck billed, platypus and spiny ant-eater, which lay eggs.
- Their bodies are hairy or covered with fur.
- Most mammals walk on their limbs (legs). A Bat flies.

- They feed on flesh or vegetables. Examples, Rodents, bats, marines, Primates.

BIRDS

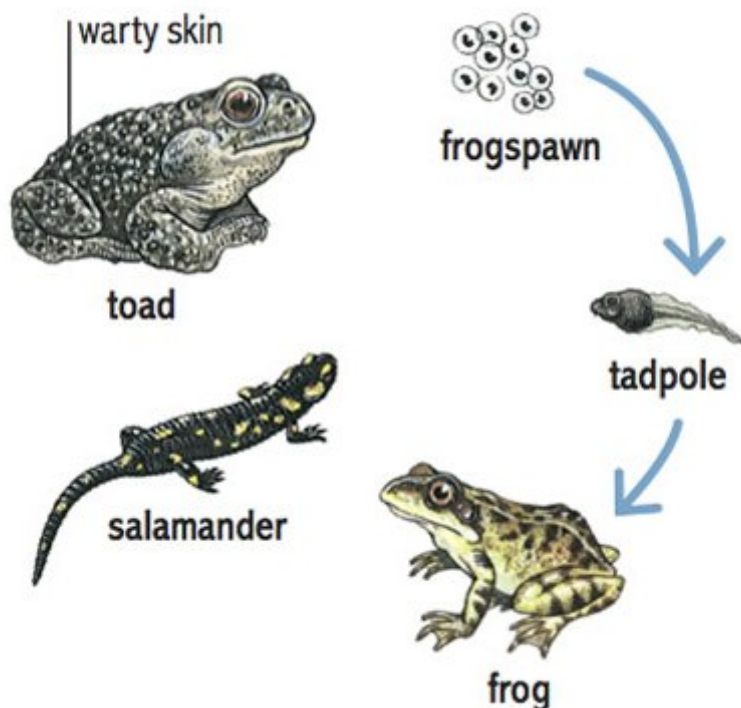
Characteristics

- Their bodies covered with feathers.
- They have beaks instead of teeth.
- They lay eggs. They incubate their eggs to hatch into a chick.
- They have wings. Most birds use them for flying.
- They have scales on their feet.
- They breathe through lungs, Examples, Nectar suckers, wading, filter feeders, birds of prey.

COLD BLOODED

- Their body temperature is not constant.
- It changes according to surrounding, examples;-

AMPHIBIANS



Characteristics

- They have moist skin without scales.
- They reproduce by laying eggs in water.

- They have external fertilization.
 - They live partly in water and partly on land.
- Young ones breath through gills while adult breath through lungs.
- They can also breathe through moist skin. Examples Frogs, Salamanders, toads, Newts.

FISH

- They have fins.
- Bodies covered with scales.
- Produce by laying eggs, which are fertilized externally.
- They breathe through gills.
- Spend their lives in water, example Shark, Tilapia, trout.

INVERTEBRATES



They are animals without backbones. Examples:

INSECTS

- e.g. houseflies, bees, butterflies, aphids, termites, fleas, lice.
- Have three body regions i.e. Head Thorax, abdomen.
 - They reproduce by laying eggs.
 - Have a pair of antennae or feelers.

WORMS

e.g earthworms, roundworms, tapeworms NB: Ringworm is a fungi not a worm.

CENTIPEDES AND MILLIPEDES

- Centipedes protect themselves by stinging.

TICKS, SCORPIONS, MITES, SPIDERS

- They have two body parts.
- They have four pairs of legs.
- They do not fly.

(f) SNAILS AND SLUGS

- Snails have shells while slugs have shells.

REPTILES

Characteristics

- They reproduce by laying eggs.
- Their bodies are covered by dry scales.
- They breathe through lungs, examples snakes, lizards, turtles and tortoises.

ANIMAL FEEDING

Animals feed to be healthy and strong for proper and maximum production.

TYPES OF ANIMAL FEEDS

- They include pastures, fodder crops and commercial feeds.

PASTURES



- It is a piece of land where grass or a mixture of legumes and grass are grown.
 - Animals feed on pasture directly.
 - In a pure stand only one type of pasture is grown.
 - Mixed pasture or mixed e.g. is where two types are planted.
 - Sometimes pasture can be harvested and fed to livestock. Examples Kikuyu grass, guinea grass, Star grass.
- Legumes include Desmodium, lucern, clover, alfalfa, glycine.

FODDER CROPS.



- They are grown harvested cut and fed to the livestock.
- Animals do not graze directly. E.g. Napier grass, kales, oats, maize, sweet potato vines, sorghum, sunflower, Guatemala grass, Rhodes grass, star grass, kikuyu grass.

CONSERVED FEEDS

- They are crops or pastures harvested and preserved for the future.
- They are preserved as silage or hay.



SILAGE

- It fodder preserved when still green in a pit or silo.
- Legumes e.g. Lucerne, glycine together with grasses or straws can be used.
- The fodder is cut into small pieces and then compressed in a silo to remove air.
- This prevents rotting.



HAY

Grasses and legumes are cut and dried.
It is sold in bales.

COMMERCIAL FEEDS

- They are also called concentrates or manufactured feeds.
- They contain high concentration of nutrients.
- They are manufactured depending on:-
 - Age of animals.
 - Type of livestock.
 - Reason for keeping the livestock.
- They are in form of Pellets, mashes, cubes.

TYPES OF CONCENTRATES

They are substances which make feeds more nutritious.

There are three types:-

(a) Carbohydrates concentrate

- They are rich source of energy.
 - They are prepared mainly from cereal by products e.g. wheat, sorghum, maize etc.
- e.g. Dairy meal, broilers mash etc.

(b) Animal Protein Concentrates

- They mainly contain body building nutrients
- They also help to repair worn out body tissues.
- Young animals require proteins to help in body building, e.g. fish meals bone meal, blood meal.

(c) Vegetable Protein Concentrates

They are main from plants e.g. Cotton seed cake, soya beans, coconut cake, groundnut cake and
Bean meals, calf pellets.

METHODS OF GRAZING

- (a) Zero grazing or stall feeding.
- (b) Rotational grazing
- (c) Herding

ROTATIONAL GRAZING

- Pasture is divided into several parts.
- Animals graze on a part of the pasture for a period.
- When grass exhausted animals are moved to next part.

METHODS OF ROTATIONAL GRAZING

(a) Paddocking



- Pastures are divided into smaller units called paddocks.
- Permanent fences are used.
- Animals are grazed in one paddock for a short time and then moved to another.
- Paddocks have watering points.
- Requires large pieces of land.

(b) Tethering

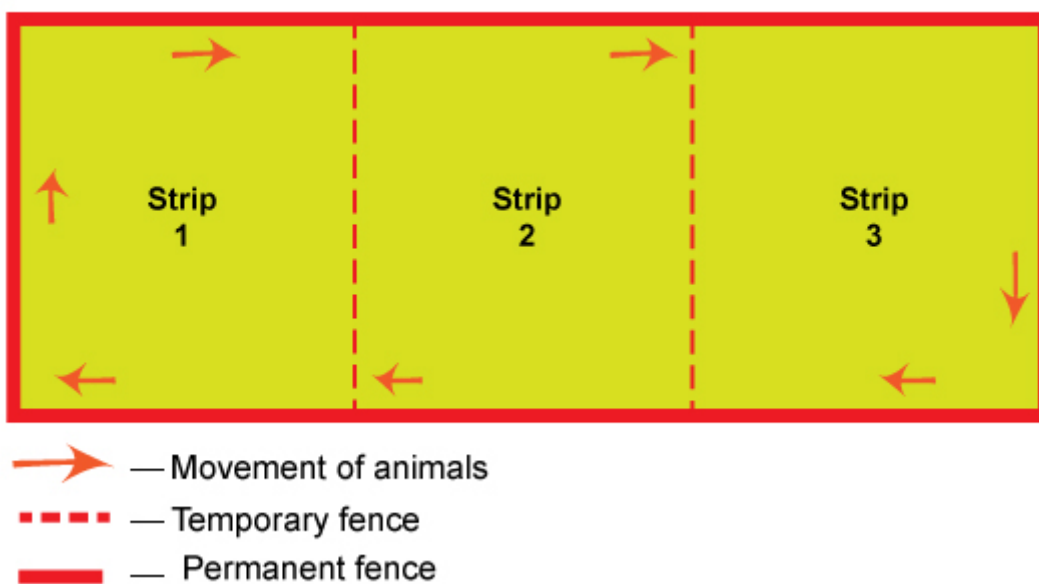


- A long rope tied to a peg is used to control the movement of the animal.
- Animal grazes on the grass within the radius of the rope.
- It is used where few animals are grazed.
- Water must be provided.

(c) Strip grazing

- Temporary or moveable fence like electric fence branches with thorns are used.

Strip grazing



ADVANTAGES OF ROTATIONAL GRAZING

- Pasture is properly used without wastage.
- Help control pests and parasites.
- Soil erosion is minimized.
- Pasture has time to regenerate or recover.
- Dung is evenly distributed in the field.
- Easy to replant dry legumes or grass.
- Easy to replant dry legumes or grass.
- Extra grass can be used as hay.

Disadvantages

- Ox in tethering animal can easily injure itself.
- It is expensive to start.
- Requires large piece of land.
- In tethering the animals must be changed frequently.

ZERO GRAZING

- Also called stall feeding.
- Animals movement is restricted.
- Water and food must be provided.

Barbed wire



Requirements for Zero grazing

- Feed trough
- Water trough
- Store to keep feed and equipment
- A shed for protection against rainfall and heat.
- Labour to feed and clean the stall.
- A garden to grow fodder.

Advantages

- Animals do not waste energy.
- High rate of production
- Less wastage of food.
- Less chances of contacting diseases.
- Easier to spot sick animal.
- Easy to keep records of each animal.
- Easy to collect manure.

Disadvantages

- Animals lack enough exercise.
- It is expensive to start.
- Requires a lot of labour.
- Requires high standard of hygiene.
- Feed must be grown or brought and taken to the animal.

HERDING

- Animals graze in large tracts of land.
- Animals are allowed to move freely without restrictions.
- A herds man or woman takes care of the animals.
- Animals select their feed.

Disadvantages

- Large pieces of land are needed.
- Parasites and diseases are easily spread.
- A lot of energy is lost.
- Animals can stray into garden destroy crops.
- They can cause erosion.

Constituents of balanced diet

Livestock requires balanced diet for

(a) Proper growth

(b) High production of meat, milk and other animal products.

(c) Protection against diseases.

(d) Warmth

A balanced diet ration should contain:-

- Needed to provide energy sources, maize germ, wheat bran, rice husks, grasses honey molasses.

PROTEINS

- Needed for growth and repair worn out tissues production of animals products.

Sources

- Legumes, e.g. lucern, desmodium, alfafa.
- Concentrates .e.g. broilers mash, growers mash, layers mash.

FATS AND OILS

Provide energy and warmth.

Sources

Cotton seed, sunflower, plant leaves, fish meal of oats.

MINERALS AND VITAMINS

Help in formation of bones and protect body against diseases.

Sources

- Bones meals, salt licks, fish meal,
- Vitamins
- Green fodder
- Fruits

ROUGHAGES

- They have nutritive value
 - They help in digestion and prevent constipation sources.
- Hay straws from rice, maize, wheat, fodder crops.

WATER

Need for (1) Cooling the body.

(2) Help in excretion.

(3) Help transport digested food in the body.

(4) Help in formation of body fluids e.g. blood.

Source

- Green fodder

LIVESTOCK PARASITES

- A parasite is an organism that lives on and gets its nourishment (food) from another living thing.
- A parasite spends part or whole of its life on a host.

They are classified into two groups.

(a) Internal parasites

(b) External parasites

- Internal parasites live inside the body of the host .e.g. liver flukes, tapeworms, roundworms.

- Internal parasites are commonly found in the liver, lungs, stomach and intestines.

PARASITE	ANIMAL ATTACKED	EFFECTS ON HOST
Liver flukes	Goats, sheep, pigs, Cattles found In liver, lungs	Make tunnels in the liver of The host.
Tapeworms	Cattle, Sheep, goats, pigs, man found in small	Absorb digested food from

	intestines	host
Round worms	Cattle, Sheep, goats, pigs, rabbits, Poultry, found in small intestines, lungs, liver	Feed on feed that the host has eaten leading to weight loss.

EXTERNAL PARASITES

PARASITE	EFFECTS OF PARASITES	ANIMAL ATTACKED
TICK	- Suck blood causing anaemia - Transmit diseases - Damage hide or the skin of the host - Cause irritation	Cattle, sheep, goats, pigs, poultry
MITES	- Suck blood causing anaemia - Cause irritation - Transmit diseases - Damages hides and skins - Cause falling off of wool in sheep	Sheep, pigs, Poultry, Cattle, Goats
TSE-TSE FLY	- Cause irritation in rabbits, horses and camels - Anaemia	All domestic animals
FLEAS	- Suck blood - Cause irritation - Spread diseases - Anaemia	Goats, Poultry, Sheep, Pigs, Rabbits
LICE	- Suck blood - Spread diseases	Horses, Pigs, Cattle, Poultry, Sheep

EFFECTS OF LIVESTOCK PARASITES

- Lower quality of animal produce .e.g. eggs, milk
- Lower quality of produce .e.g. amount of milk.
- Cause irritation
- Transmit disease e.g. East Coast Fever.
- They cause anaemia.
- They damage skins and hides
- Weakens animals.
- Animals are usually hungry.

WAYS OF CONTROLLING IN LIVESTOCK PARASITES

- Dipping to kill external parasites.
- Spraying to kill external parasites. Acancides are used.
- Rotational grazing for both internal and external parasites.
- Deworming to control internal parasites.
- Hand picking to control internal parasites.
- Culling – Removal of infected and unproductive animals from the stock .

SIGNS OF ILL-HEALTH IN LIVESTOCK

- Stunted growth
- Loss of weight
- Reduce yields
- Coughing
- Blood in stool or urine
- Rough coat or fur
- Dry muzzle

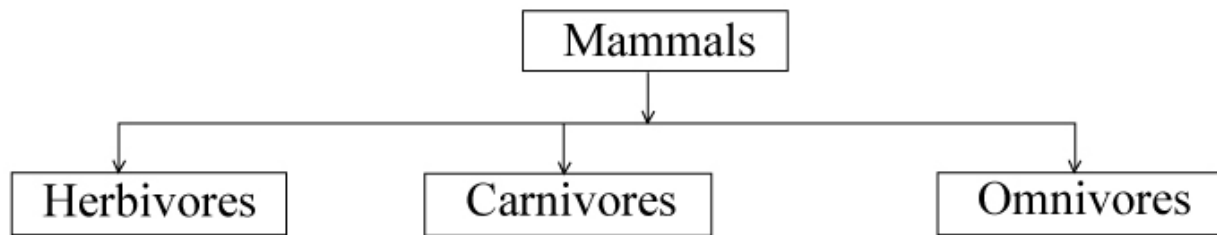
EFFECTS OF LIVESTOCK DISEASES

- Low yields
- Reduced quality of the produce
- Transmission of diseases
- Death of the animals.
- General economic losses.

ADAPTATION OF ANIMALS

Adaptations are features behaviors and characteristics of an animal that enables it to survive in a particular environment.

- (i) Adaptations to feeding habits.
- (ii) Adaptation for movement.

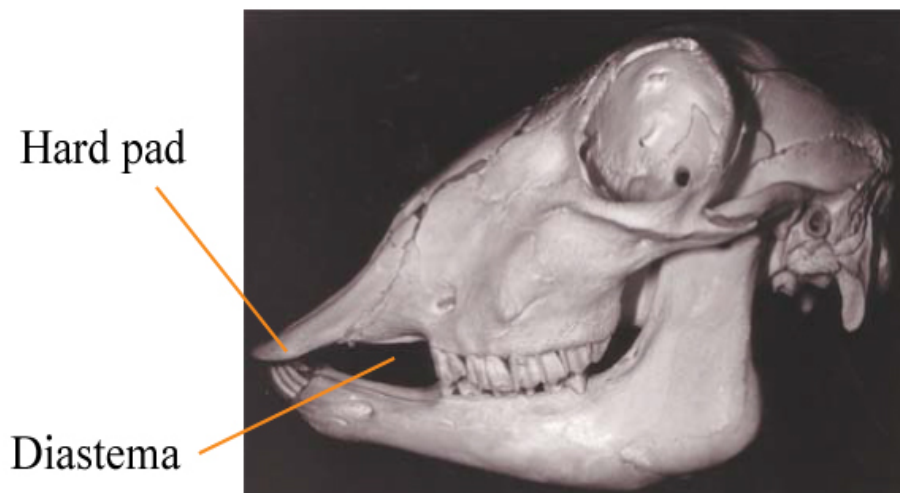


FEEDING HABITS

(a) Herbivores

- They are animals that feed on plants.-
- Some herbivorous graze and other browse on bushes and shrubs.-
- Grazers feed from the ground e.g. cows, sheep, horses, browsers feed on twigs .e.g. giraffes, goats etc.

Diagram of a herbivorous animal jaw



(b) Ruminants

- They have four stomach chambers.
- They chew cud.
- They have a toothless gap between incisors and premolars on the lower jaw called distema.
- Distema help them to turn food easily while chewing.
- Horny pad – no incisors and canines present in the upper jaw. It helps to hold the grass firmly.
- Rough tongue It protects the animal from injury when feeding.

(c) Carnivores

- Feed on flesh of other animals .e.g. lions,

Incisors

- They are small pointed in both jaws.
- Used to grip and strip off small pieces of flesh from bones.

- Canines

Shapes like a dagger.

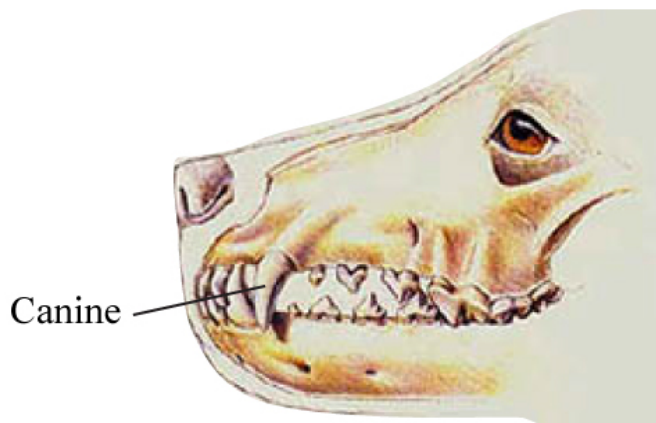
Used for tearing off pieces of flesh from bones.

They are also to hold and kill its prey.

-Pre-molars and Molars

- Have flat top for crashing and modified into a special teeth called Carnassials teeth.
- They act like scissors slicing for flesh and crack bones.

Diagram for Carnivores jaw.



(e) Omnivores- They feed on both plants and flesh .e.g. man, monkey

Incisors- Shapes like chisel used for biting and cut off pieces of food.

Canines- Sharp and pointed for tearing.

Pre-molars and molars - They have ridges known as cusps used for grinding food.

BIRDS

ADAPTATIONS TO FEEDING: BEAKS OF BIRDS

- Birds have beaks instead of teeth.
- We can tell what kind of food a feeds on by observing its beak.

1. Grain eating birds

- They have short, Strong; Conical and pointed blunt beaks e.g. Chicken, Turkey.



(ii) Their beaks (bills) are

- Conical
- Short
- Straight
- Strong
- Thick

2. Flesh eaters(birds of prey)



- Feed entirely on flesh
- They have: sharp strong hooked (curved) beaks
- They use their beaks for tearing flesh or meat from their bones of their prey

3. Nectar feeders



- Their beaks are long, slender (thin) that are slightly curved.
- They use their beaks to reach for nectar at the bottom of flowers.

4. Filter feeder

.e.g Flamingoes, ducks.



- Their beaks are flat broad and notched (serrated) with cross-plates.
- They used for scooping and straining their catch.
- They sieve their food out of muddy water.

ADAPTATION TO MOVEMENT

Movement is also called locomotion.

FLIGHT ADAPTATIONS

Birds move by flying. Easily due to

a) Streamlined bodies

- This helps to reduce friction or air resistance.

- b) They have wings – feathers attached to wings form continuers air proof surface that help to be supported by air pressure while flying.
 - c) Large powerful muscles to power wings during flight.
 - d) Hollow and light bones which reduce total weight of the bird.
 - e) A rigid skeleton which help in attachment of flight muscles.
- N.B Feathers are smooth and lie backwards to ease flow of air when flying.

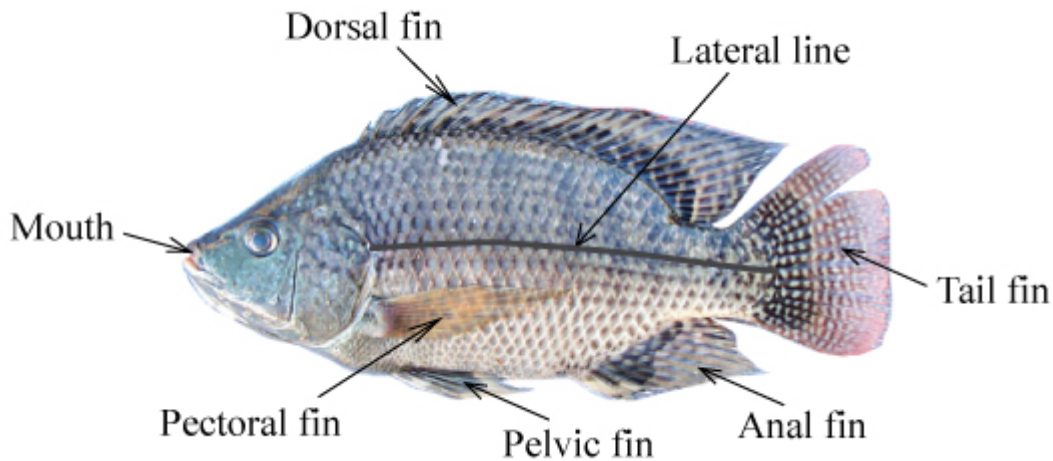
ADAPTATIONS OF ANIMALS THAT MOVE IN WATER

Swimming

Fish, it has:

- i. Streamlined body – This helps to reduce friction during movements.
- ii. The body is slimy (slippery) – To reduce friction.
- iii. Pressure of fins to propel and balance during motion.
- iv. Strong muscles – To produce force needed for movement.
- v. Overlapping scales – To reduce friction.

NB – Fish has air bladder to help c ontrol its floating angle when not swimming.



- Other Swimming animals have webbed feet .e.g. duck and frogs.
- Turtles have flippers instead of feet to help them in swimming.

Hopping and leaping

- Such animals have long and strong hind limbs .e.g. frogs, toads, grasshoppers.
- They have short – fore limbs that enables them to land.